When friction converts rocking energy to heat, how hot does the planet's interior get? Hot enough to trigger volcanos and lava flows?

Jupiter's moon Io is a remarkable example of this. Io, the large moon that orbits closest to Jupiter's surface, doesn't rock. But it does move closer and farther from Jupiter along an elliptical orbit, so it feels Jupiter's tidal gravity strengthen then weaken then strengthen, much like Miller's planet feels Gargantua's tidal gravity oscillate. This heats Io enough to produce huge volcanos and lava flows (Figure 17.8).

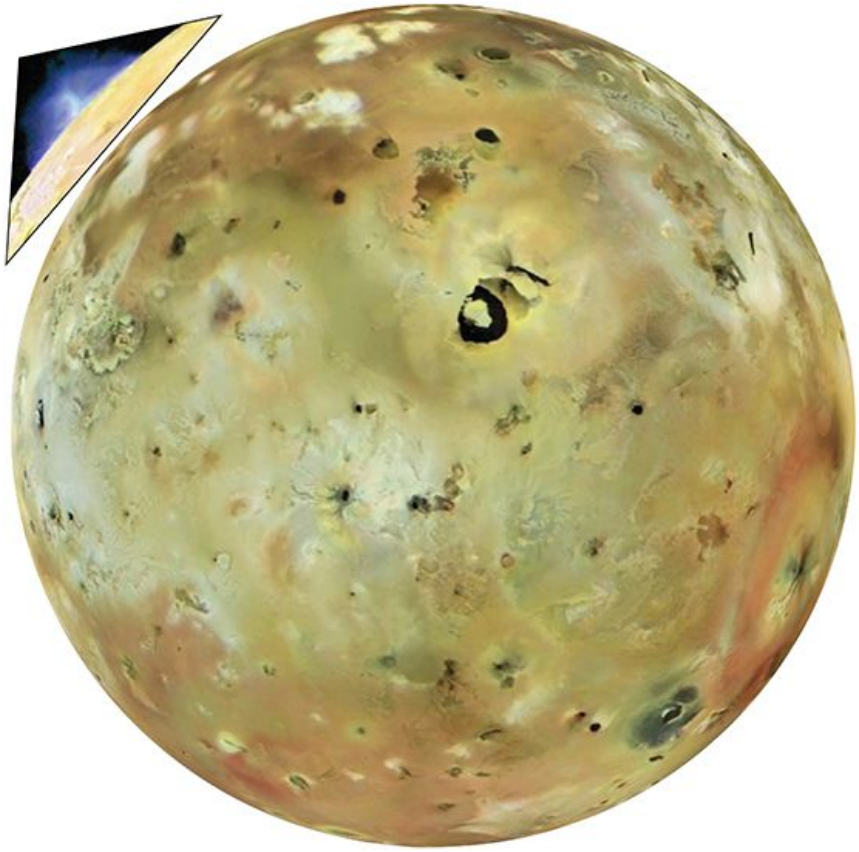


Fig. 17.8. Io as photographed by the *Galileo* spacecraft shows many volcanos and lava flows. *Inset*: A 50-kilometer-high volcanic plume.

The Appearance of Gargantua from Miller's Planet